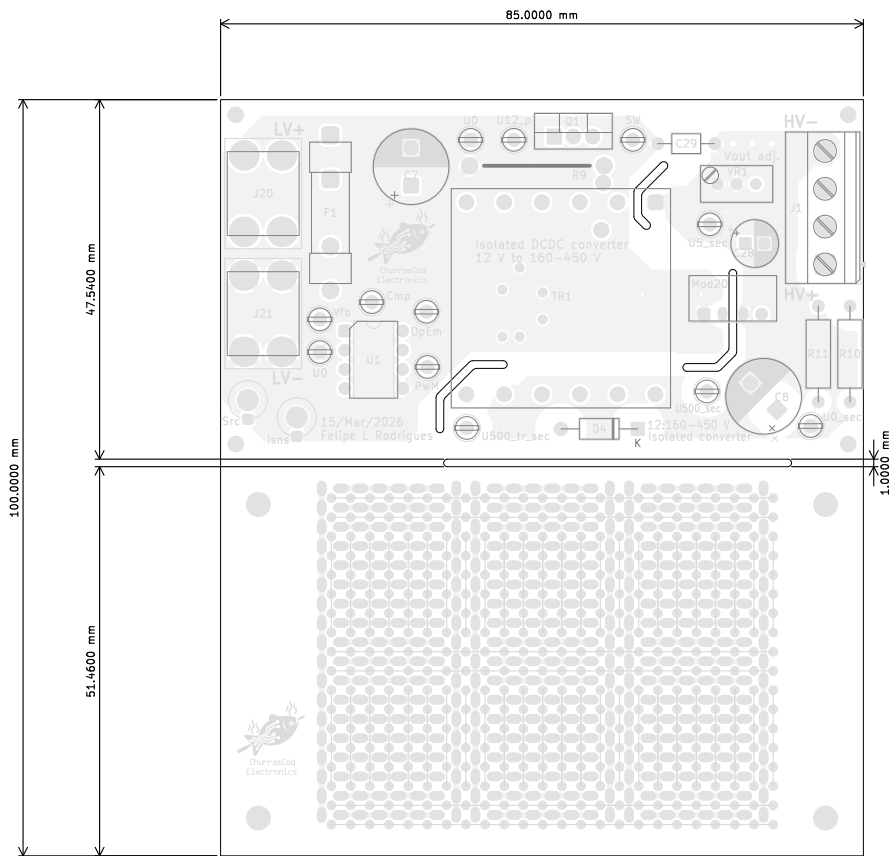


Step-Up Fabrication Document

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

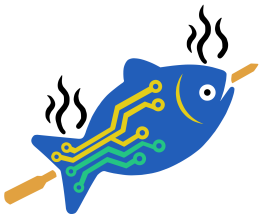
- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

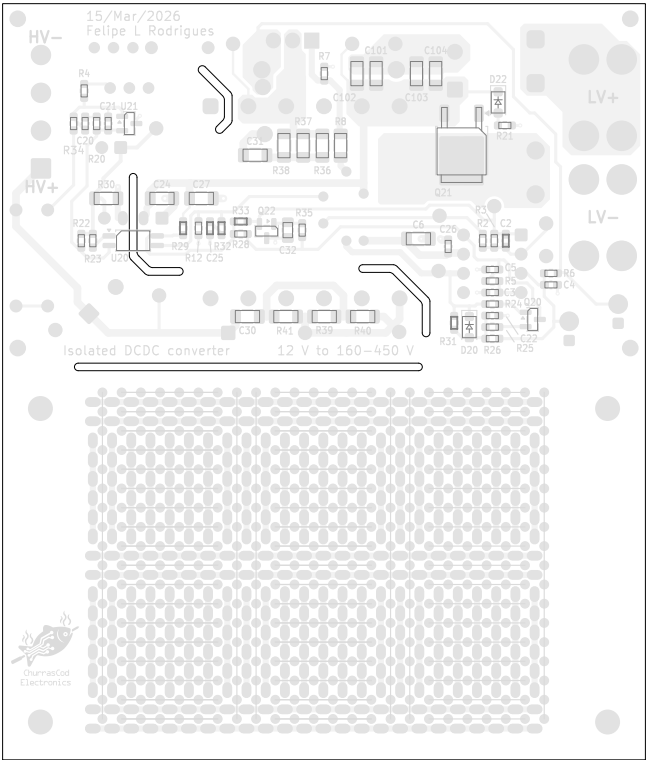
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NA.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE 85,000 × 100,000 mm
BOARD THICKNESS 1.602 mm
TRACE WIDTH 0.300 mm
TRACE TO TRACE -0,000 mm
MIN. HOLE (PTH) 0,250 mm
MIN. HOLE (NPTH) 2,100 mm
ANNULAR RING 0,150 mm
COPPER TO HOLE 0,200 mm
COPPER TO EDGE 0,250 mm
HOLE TO HOLE 0,254 mm



ChurrasCod
Electronics

Comments:	Company: NA		Variant: CHECKED		Git Hash: 7f19023	
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V			
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb		Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)	
Sheet Path:			Reviewer: FR	Size: A4	Sheet: 1 of 7	



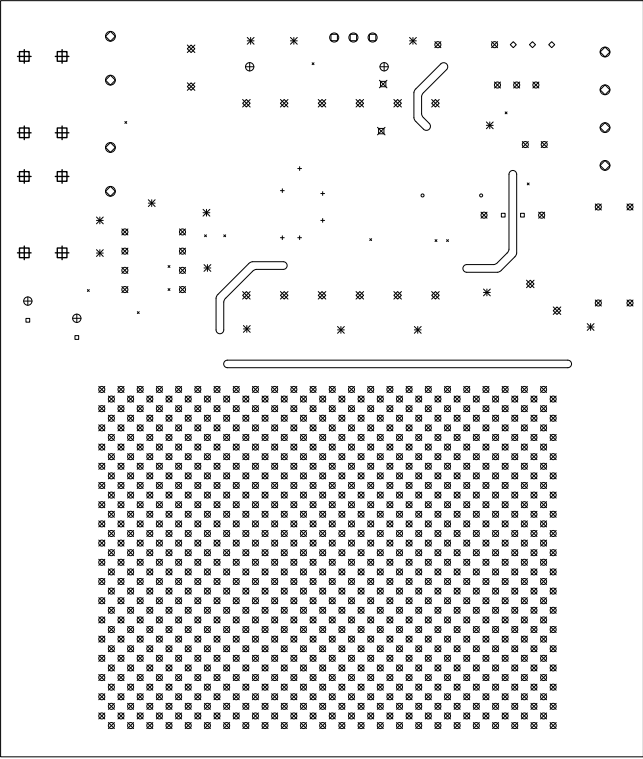
Comments: Sheet Title: Bottom Fabrication (Scale 1:1) Sheet Path:	Company: NA		Variant: CHECKED		Git Hash: 7f19023
	Board Name: Step-Up			Project Name: Isolated step-up 12:450 V	
	File Name: Isolated_12to450V.kicad_pcb		Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
	Reviewer: FR		Size: A4	Sheet: 2 of 7	

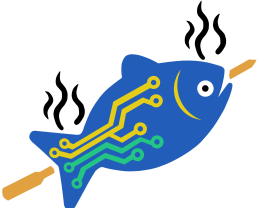
Step-Up Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
○	2	0.40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
+	6	0.50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
□	4	0.70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
◇	3	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
⊗	885	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
※	14	0.90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊗	2	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
⊗	16	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	4	1.10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	3	1.20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
○	8	1.30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	8	1.80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
Total 968						





ChurrasCod
Electronics

Comments:
Drill Drawing (L1 - L4)

Sheet Title:
Drill Drawing (L1 - L4)

Sheet Path:

Company:
NA

Board Name:
Step-Up

File Name:
Isolated_12to450V.kicad_pcb

Designer:
FR

Reviewer:
FR

Variant:
CHECKED

Git Hash:
7f19023

Project Name:
Isolated step-up 12:450 V

Date:
2026-02-07

Revision:
2.0.0+ (Unreleased)

Size:
A4

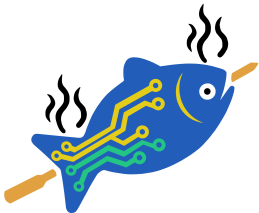
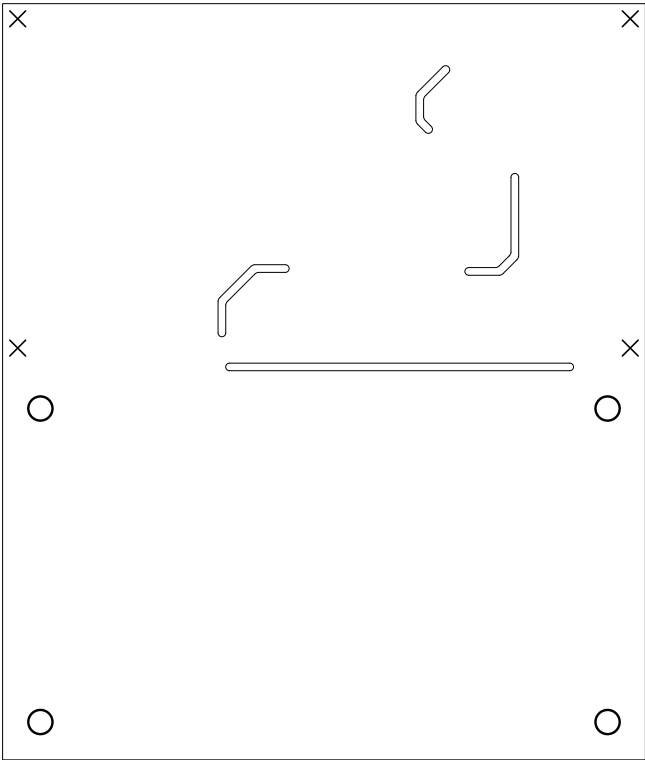
Sheet:
3 of 7

Step-Up Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
○	4	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
Total 8						



ChurrasCod
Electronics

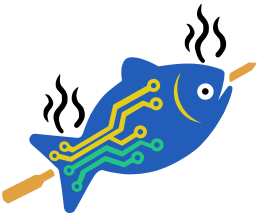
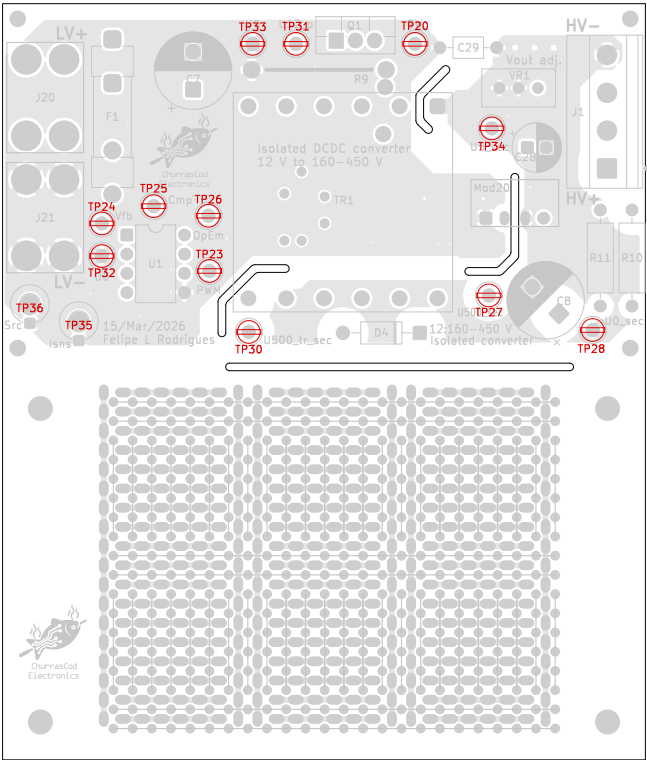
Comments:	Company: NA		Variant: CHECKED	Git Hash: 7f19023	
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V		
Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)	
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 4 of 7	

Step-Up Fabrication Document

Top Test Points (Scale 1:1)

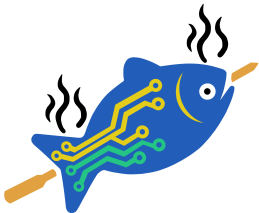
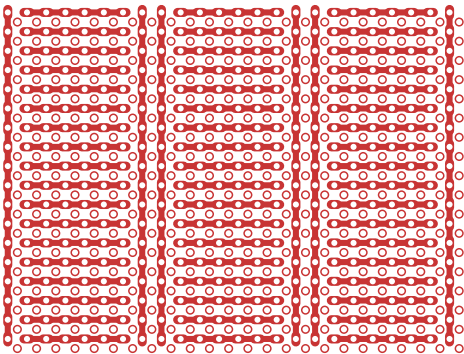
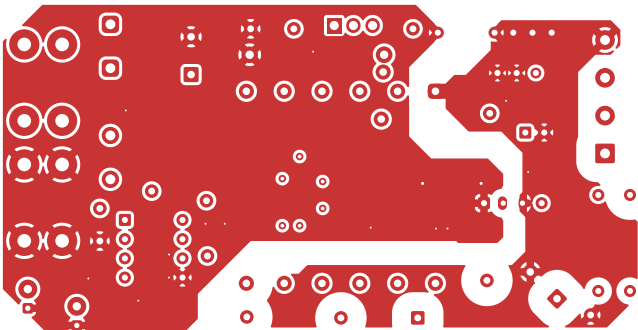
Ref.	Net	X [mm]	Y [mm]
TP20	SW	56.53	44.66
TP23	PWM	29.35	14.62
TP24	Vfb	15.13	20.91
TP25	Comp	21.98	23.20
TP26	OpEm	29.22	21.93
TP27	U500_sec	66.31	11.39
TP28	U0_sec	80.02	6.81
TP30	U500_tr_sec	34.56	6.56
TP31	U12_p	40.78	44.66
TP32	U0	15.13	16.59
TP33	U0	35.06	44.66
TP34	U5_sec	66.69	33.48
TP35	Isns,U0	12.08	6.70
TP36	Src	5.60	8.97
TP101	Net-(TP101-Pad1)	17.92	-1.44
TP102	Net-(TP101-Pad1)	20.46	-1.44
TP103	Net-(TP101-Pad1)	22.54	-1.44
TP104	Net-(TP101-Pad1)	25.54	-1.44
TP105	Net-(TP101-Pad1)	28.08	-1.44
TP106	Net-(TP101-Pad1)	30.62	-1.44
TP107	Net-(TP107-Pad1)	17.92	-3.98
TP108	Net-(TP107-Pad1)	20.46	-3.98
TP109	Net-(TP107-Pad1)	23.00	-3.98
TP110	Net-(TP107-Pad1)	25.54	-3.98
TP111	Net-(TP107-Pad1)	28.08	-3.98
TP112	Net-(TP107-Pad1)	30.62	-3.98
TP113	Net-(TP113-Pad1)	17.92	-6.52
TP114	Net-(TP113-Pad1)	20.46	-6.52
TP115	Net-(TP113-Pad1)	23.00	-6.52
TP116	Net-(TP113-Pad1)	25.54	-6.52
TP117	Net-(TP113-Pad1)	28.08	-6.52
TP118	Net-(TP113-Pad1)	30.62	-6.52

All dimensions are in millimeters unless otherwise specified.



ChurrasCod
Electronics

Comments:	Company:		Variant:	Git Hash:
	NA		CHECKED	7f19023
Sheet Title:	Board Name:		Project Name:	
	Top Test Points (Scale 1:1)		Isolated step-up 12:450 V	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)
		Reviewer:	Size:	Sheet:
		FR	A4	5 of 7

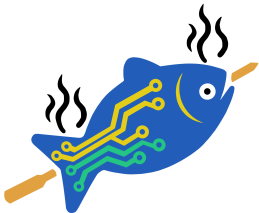
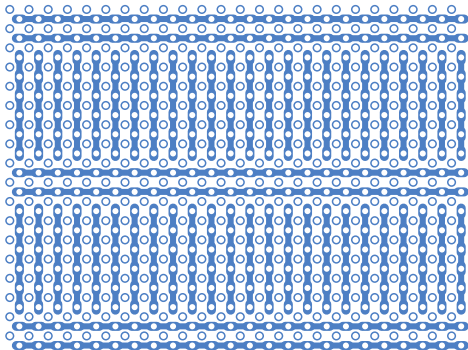
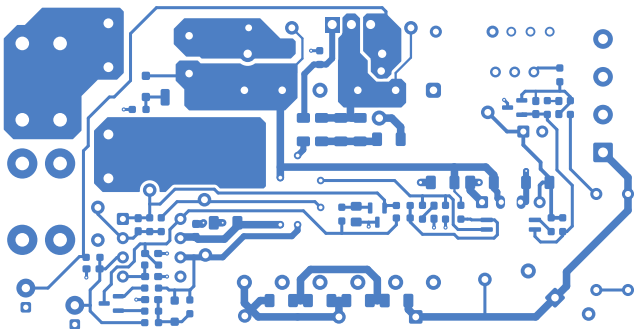


ChurrasCod
Electronics

Comments:	Company: NA		Variant: CHECKED	Git Hash: 7f19023
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
	Sheet Title: F,Cu (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 6 of 7

Step-Up Fabrication Document

B.Cu (Scale 1:1)



ChurrasCod
Electronics

Comments:	Company:		Variant:	Git Hash:
	NA		CHECKED	7f19023
	Board Name:		Project Name:	
Sheet Title:	Step-Up		Isolated step-up 12:450 V	
	File Name:	Designer:	Date:	Revision:
B.Cu (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size:	Sheet:
		FR	A4	7 of 7